THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS CHAPTER TEST SET B

Chapter 3: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The Kaprekar constant for 4-digit numbers is: 1

(a) 1089 **(b)** 6174 **(c)** 1729 **(d)** 9999

2. Starting at 6, the Collatz sequence is 6, 3, 10, 5, 16, 8, 4, 2, 1. The number of steps is: 1

(a) 9 **(b)** 8 **(c)** 6 **(d)** 10

3. The sum of the digits of 4567 is: 1

(a) 20 **(b)** 23 **(c)** 22 **(d)** 21

4. Rounding 6743 to the nearest thousand gives: 1

(a) 7000 **(b)** 6800 **(c)** 6000 **(d)** 6700

5. On a number line, numbers increase: 1

(a) from left to right **(b)** from right to left
(c) randomly **(d)** up to down only

6. The number 1089 magic: 3-digit number minus its reverse, then add the reverse of the result. The final answer is: 1

(a) 1089 **(b)** 1111 **(c)** 1001 **(d)** 999

7. The number 1001 is a palindrome of how many digits? 1

(a) 2 **(b)** 5 **(c)** 3 **(d)** 4

8. A number ending in 5 is: 1

(a) divisible by 2 **(b)** divisible by 5 **(c)** always prime **(d)** always even

9. The Collatz sequence starting at 12 takes how many steps to reach 1? 1

(a) 8 **(b)** 9 **(c)** 10 **(d)** 12

10. One step of the Kaprekar routine on 5231 gives: 1

(a) 4176 **(b)** 4086 **(c)** 3996 **(d)** 4095

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Estimate 4,876 + 2,123 by rounding to the nearest thousand. 2

12. What is the Collatz conjecture? 2

13. Is the sum of two odd numbers odd or even? Explain with an example. 2

14. Round 7,468 to the nearest hundred and to the nearest thousand. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Describe the 1089 trick with the number 532. 3

16. How many days from 10 Feb 2025 to 10 May 2025? Show working. 3

17. Two players alternately take 1, 2 or 3 sticks from a pile of 21; whoever takes the last stick wins. Describe a winning strategy for the first player. 3

OR

Round 548,123 to the nearest thousand and to the nearest ten thousand. 3

18. What are the largest and smallest 5-digit numbers formed by 7, 0, 3, 9, 1 without repeating a digit? Find their difference. 3

OR

How do you check whether a 3×3 grid of 1 to 9 is a magic square? Which number must be in the centre if every line sums to 15? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Estimate 3,987 + 5,012 + 1,008. 5
(ii) Estimate 49 × 21.
(iii) Compare with exact values.

OR

(i) Describe the 1089 trick. 5
(ii) Check it for 721.
(iii) Explain what goes wrong if the digits are all the same.

20. **Case study:** A calendar shows dates in a 7-column grid. 5
(i) Pick a 2×2 box 8, 9, 15, 16 and find the sum of diagonals.
(ii) What do you notice?
(iii) Test with another box.
(iv) Explain why.

OR

(i) Explain mental maths strategies for addition. 5
(ii) Compute 99 + 76 mentally.
(iii) Compute 5 × 48.

END OF QUESTION PAPER — Total: 40 Marks

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